

6. Density theorems for Hausdorff and packing measures

One of the most important single results for the Lebesgue measure is the Lebesgue density theorem. Here we shall first look at how much of it can be extended to s -dimensional Hausdorff measures in $\mathbf{R}^n$. Then we shall study similar questions for packing measures.

Density estimates for Hausdorff measures

We first define the spherical densities for the Hausdorff measures.

6.1. Definition. Let $0 \leq s < \infty$, $A \subset \mathbf{R}^n$ and $a \in \mathbf{R}^n$. The upper and lower s -densities of A at a are defined by

$$\begin{aligned}\Theta^{*s}(A, a) &= \limsup_{r \downarrow 0} (2r)^{-s} \mathcal{H}^s(A \cap B(a, r)), \\ \Theta_*^s(A, a) &= \liminf_{r \downarrow 0} (2r)^{-s} \mathcal{H}^s(A \cap B(a, r)).\end{aligned}$$

If they agree, their common value is called the s -dimensional density of A at a and denoted by

$$\Theta^s(A, a) = \Theta^{*s}(A, a) = \Theta_*^s(A, a).$$

In the case $s = n$, these are the usual Lebesgue densities, and the Lebesgue density theorem (recall Corollary 2.14) tells us that $\Theta^n(A, a) = 1$ for $\mathcal{L}^n$ almost all $a \in A$ and, provided A is $\mathcal{L}^n$ measurable, $\Theta^n(A, a) = 0$ for $\mathcal{L}^n$ almost all $a \in \mathbf{R}^n \setminus A$. In general, we can say much less for Hausdorff measures. However, the following theorem is often a very useful substitute for studying local properties of sets with positive and finite s -dimensional Hausdorff measure.

6.2. Theorem. Suppose $A \subset \mathbf{R}^n$ with $\mathcal{H}^s(A) < \infty$.

- (1) $2^{-s} \leq \Theta^{*s}(A, x) \leq 1$ for $\mathcal{H}^s$ almost all $x \in A$.
- (2) If A is $\mathcal{H}^s$ measurable, $\Theta^{*s}(A, x) = 0$ for $\mathcal{H}^s$ almost all $x \in \mathbf{R}^n \setminus A$.

Proof. Let us first prove the left hand inequality of (1). The set of those $x \in A$ for which $\Theta^{*s}(A, x) < 2^{-s}$ is the union of the sets

$$\begin{aligned}B_k &= \{x \in A : \mathcal{H}^s(A \cap B(x, r)) < (k/(k+1))r^s \text{ for } 0 < r < 1/k\}, \\ &\hspace{25em} k = 1, 2, \dots\end{aligned}$$

Hence it suffices to show that $\mathcal{H}^s(B_k) = 0$ for every k . Fix k , put $t = k/(k+1)$, and let $\varepsilon > 0$. We can cover B_k by sets $E_1, E_2, \dots$ such that $0 < d(E_i) < 1/k$, $B_k \cap E_i \neq \emptyset$, and

$$\sum_i d(E_i)^s \leq \mathcal{H}^s(B_k) + \varepsilon.$$

For each i pick $x_i \in B_k \cap E_i$ and let $r_i = d(E_i)$. Then $B_k \cap E_i \subset A \cap B(x_i, r_i)$ and

$$\begin{aligned} \mathcal{H}^s(B_k) &\leq \sum_i \mathcal{H}^s(B_k \cap E_i) \leq \sum_i \mathcal{H}^s(A \cap B(x_i, r_i)) \\ &< \sum_i t r_i^s = t \sum_i d(E_i)^s \leq t(\mathcal{H}^s(B_k) + \varepsilon). \end{aligned}$$

Letting $\varepsilon \downarrow 0$, we get $\mathcal{H}^s(B_k) \leq t\mathcal{H}^s(B_k)$. Since $\mathcal{H}^s(B_k) < \infty$ and $t < 1$, we have $\mathcal{H}^s(B_k) = 0$.

To prove the right hand inequality in (1) we first observe that we may assume A to be a Borel set because of the Borel regularity of $\mathcal{H}^s$. As above, letting $t > 1$ and setting

$$B = \{x \in A : \Theta^{*s}(A, x) > t\},$$

it suffices to show $\mathcal{H}^s(B) = 0$. Let $\varepsilon > 0$ and $\delta > 0$. Applying Theorem 1.10 (2) to the restriction $\mathcal{H}^s \llcorner A$, we find an open set U such that $B \subset U$ and $\mathcal{H}^s(A \cap U) < \mathcal{H}^s(B) + \varepsilon$. For every $x \in B$ there are arbitrarily small numbers r such that $0 < r < \delta/2$, $B(x, r) \subset U$ and

$$\mathcal{H}^s(A \cap B(x, r)) > t(2r)^s.$$

By Theorem 2.8 there are such disjoint balls $B_1, B_2, \dots$ for which $\mathcal{H}^s(B \setminus \bigcup_i B_i) = 0$. Thus

$$\begin{aligned} \mathcal{H}^s(B) + \varepsilon &> \mathcal{H}^s(A \cap U) \geq \sum_i \mathcal{H}^s(A \cap B_i) \\ &> t \sum_i d(B_i)^s \geq t\mathcal{H}_\delta^s\left(B \cap \bigcup_i B_i\right) = t\mathcal{H}_\delta^s(B). \end{aligned}$$

The last equality follows from $\mathcal{H}_\delta^s(B \setminus \bigcup_i B_i) = 0$ and the subadditivity of $\mathcal{H}_\delta^s$. Letting $\varepsilon \downarrow 0$ and $\delta \downarrow 0$ and using $t > 1$, we get $\mathcal{H}^s(B) = 0$.

To prove (2) we show that for any $t > 0$ the set

$$B = \{x \in \mathbf{R}^n \setminus A : \Theta^{*s}(A, x) > t\}$$

has zero $\mathcal{H}^s$ measure. Let $\varepsilon > 0$. Since $(\mathcal{H}^s \llcorner A)(B) = 0$, we find by Theorem 1.10 (2) an open set U such that $B \subset U$ and $\mathcal{H}^s(A \cap U) < \varepsilon$. For every $x \in B$ there is $r(x) > 0$ such that $B(x, r(x)) \subset U$ and

$$\mathcal{H}^s(A \cap B(x, r(x))) > t(2r(x))^s.$$

By Theorem 2.1 there are $x_1, x_2, \dots \in B$ such that the balls $B_i = B(x_i, r(x_i))$ are disjoint and the balls $5B_i$ cover B . Then

$$\begin{aligned} t\mathcal{H}_\infty^s(B) &\leq t \sum_i d(5B_i)^s = t5^s \sum_i d(B_i)^s \\ &< 5^s \sum_i \mathcal{H}^s(A \cap B_i) \leq 5^s \mathcal{H}^s(A \cap U) < 5^s \varepsilon. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ we get $\mathcal{H}_\infty^s(B) = 0$, whence $\mathcal{H}^s(B) = 0$ by Lemma 4.6. $\square$

As a corollary we obtain that the values of the densities are preserved almost everywhere in measurable subsets.

6.3. Corollary. *Let A and B be $\mathcal{H}^s$ measurable subsets of $\mathbf{R}^n$ with $B \subset A$ and $\mathcal{H}^s(A) < \infty$. Then for $\mathcal{H}^s$ almost all $x \in B$,*

$$\Theta^{*s}(B, x) = \Theta^{*s}(A, x) \quad \text{and} \quad \Theta_*^s(B, x) = \Theta_*^s(A, x).$$

Proof. Apply (2) to $A \setminus B$. $\square$

6.4. Remarks. (1) Theorem 6.2 is essentially due to Besicovitch [1]. Because of it the upper densities are in general more useful than the lower densities. It may well happen that a compact set with positive and finite $\mathcal{H}^s$ measure has zero lower density at every point. For example, consider the very porous sets discussed at the end of 4.12.

(2) The upper bound in 6.2 (1) is always sharp and the lower bound 2^{-s} is the best possible for $s \leq 1$. It may not be the best possible for $s > 1$; in fact, it is not known if 2^{-s} could always be replaced by $1/2$. For some values of s M. Chlebík (unpublished) has found better lower bounds for the spherical measure.

(3) There are many closely related and more general density theorems. For example, proving first a covering theorem with more general sets than balls, one can show with similar arguments that if $\mathcal{H}^s(A) < \infty$, then

$$\limsup_{\delta \downarrow 0} \{ \mathcal{H}^s(A \cap E)/d(E)^s : x \in E, 0 < d(E) < \delta \} = 1$$

for $\mathcal{H}^s$ almost all $x \in A$, see Falconer [4, Theorem 2.3] or Federer [3, 2.10.17]. This generalizes 6.2(1). Below in Theorem 6.6 we shall prove the corresponding result for spherical measures. For more general density theorems of the above type applying to the generalized Hausdorff measures Λ_h and many others, see Federer [3, 2.10], Davies and Samuels [1] and H. W. Pu and H. H. Pu [1]. Some other local questions related to Hausdorff measures are treated in Barlow and Taylor [1] and Kirchheim [2].

(4) Theorem 6.2 holds also if the densities are defined in terms of $\mathcal{H}_\infty^s$ in place of $\mathcal{H}^s$. Of course, this is an improvement only in the case of the lower bound 2^{-s} . The proof is the same. For other density and covering results for $\mathcal{H}_\infty^s$, see Fernström [1], Mateu and Orobitg [1], Mattila and Orobitg [1], Melnikov and Orobitg [1] and O'Farrell [1].

(5) As noted in Theorem 4.14 more can be said about the densities of the self-similar K with the open set condition. In particular, $0 < a \leq \Theta_*^s(K, x) \leq \Theta^{*s}(K, x) \leq b < \infty$ for $x \in K$. Moreover, Salli [1] proved that for some c ,

$$\Theta^{*s}(K, x) = c \quad \text{for } \mathcal{H}^s \text{ almost all } x \in K,$$

and the analogous statement holds for the lower densities provided the different parts $S_i(K)$ are disjoint. This is essentially an ergodicity result on the similitudes generating K . In fact, Salli proved his results for much more general conical densities instead of the spherical ones.

Compact sets E satisfying (like the self-similar sets with the open set condition)

$$0 < ar^s \leq \mathcal{H}^s(E \cap B(x, r)) \leq br^s < \infty \quad \text{for } x \in E, 0 < r \leq 1,$$

have recently been investigated in different situations. For example, Jonsson and Wallin [1] have used them in connection with some function spaces, David, Semmes and others have studied them together with some problems in complex and harmonic analysis, see Christ [1], David [4] and David and Semmes [1]–[2]. They are often called regular or Ahlfors–David regular.

A density theorem for spherical measures

Recall the definition of the spherical measure $\mathcal{S}^s$ from 5.1.

6.5. Definition. For $A \subset \mathbf{R}^n$ and $a \in \mathbf{R}^n$ set

$$\sigma^s(A, a) = \limsup_{\delta \downarrow 0} \left\{ \frac{\mathcal{S}^s(A \cap B)}{d(B)^s} : B \text{ is a closed ball with } a \in B \text{ and } d(B) < \delta \right\}.$$

6.6. Theorem. If $A \subset \mathbf{R}^n$ with $\mathcal{S}^s(A) < \infty$, then $\sigma^s(A, x) = 1$ for $\mathcal{S}^s$ almost all $x \in A$.

Proof. The proof of $\sigma^s(A, x) \geq 1$ is similar to that of $\Theta^{*s}(A, x) \geq 2^{-s}$ in 6.2 and is left as an exercise. As in the proof of $\Theta^{*s}(A, x) \leq 1$, we may assume A to be a Borel set. Let $1 < t < 1 + 5^{-s-1}$ and

$$B = \{x \in A : \sigma^s(A, x) > t\}.$$

We have to show $\mathcal{S}^s(B) = 0$. Assume $\mathcal{S}^s(B) > 0$ and again as in the proof of 6.2 we can find an open set U containing B such that

$$(1) \quad \mathcal{S}^s(A \cap U) < t^{1/3} \mathcal{S}^s(B) < (1 + 5^{-s-1}) \mathcal{S}^s(B).$$

By the definition of $\mathcal{S}^s$ there is $\delta > 0$ such that

$$(2) \quad \sum_i d(B_i)^s > t^{-1/3} \mathcal{S}^s(B) > \frac{1}{2} \mathcal{S}^s(B)$$

whenever $B_1, B_2, \dots$ are closed balls with $d(B_i) < \delta$ covering $\mathcal{S}^s$ almost all of B . Using the covering theorem 2.1 and the definition of B , we find disjoint closed balls $B_1, B_2, \dots$ such that $B_i \subset U$, $d(B_i) < \delta/5$, $B \subset \bigcup_i 5B_i$ and

$$(3) \quad \mathcal{S}^s(A \cap B_i) > t d(B_i)^s.$$

Then by (2),

$$\begin{aligned} \mathcal{S}^s\left(A \cap \bigcup_i B_i\right) &= \sum_i \mathcal{S}^s(A \cap B_i) > \sum_i d(B_i)^s \\ &= 5^{-s} \sum_i d(5B_i)^s > 5^{-s} 2^{-1} \mathcal{S}^s(B). \end{aligned}$$

Hence for some k_1 ,

$$\mathcal{S}^s\left(A \cap \bigcup_{i=1}^{k_1} B_i\right) > 5^{-s} 2^{-1} \mathcal{S}^s(B).$$

Set $c = 1 - 5^{-s}4^{-1}$ and $C_1 = A \cap \bigcup_{i=1}^{k_1} B_i$. Then by (1)

$$\begin{aligned} \mathcal{S}^s(B \setminus C_1) &\leq \mathcal{S}^s((A \setminus C_1) \cap U) \\ &\leq \mathcal{S}^s(A \cap U) - 5^{-s}2^{-1}\mathcal{S}^s(B) \leq c\mathcal{S}^s(B). \end{aligned}$$

If $\mathcal{S}^s(B \setminus C_1) > 0$, we can repeat the same argument to find disjoint balls $B_{k_1+1}, \dots, B_{k_2}$ contained in $U \setminus \bigcup_{i=1}^{k_1} B_i$ such that $d(B_i) < \delta/5$, (3) holds and for $C_2 = A \cap \bigcup_{i=1}^{k_2} B_i$,

$$\mathcal{S}^s(B \setminus C_2) \leq c\mathcal{S}^s(B \setminus C_1) \leq c^2\mathcal{S}^s(B).$$

Continuing this we obtain a finite or countable sequence of disjoint closed balls $B_1, B_2, \dots$ with $d(B_i) < \delta$, $B_i \subset U$, satisfying (3) and

$$(4) \quad \mathcal{S}^s\left(B \setminus \bigcup_i B_i\right) = 0.$$

Then by (4), (2), (3) and (1),

$$\begin{aligned} t^{-1/3}\mathcal{S}^s(B) &< \sum_i d(B_i)^s < t^{-1} \sum_i \mathcal{S}^s(A \cap B_i) \\ &\leq t^{-1}\mathcal{S}^s(A \cap U) < t^{-2/3}\mathcal{S}^s(B), \end{aligned}$$

whence $t^{1/3}\mathcal{S}^s(B) < \mathcal{S}^s(B)$. As $t > 1$, this is a contradiction. $\square$

Since $\mathcal{H}^s \leq \mathcal{S}^s \leq 2^s\mathcal{H}^s$ we have immediately

6.7. Corollary. *If $A \subset \mathbf{R}^n$ with $\mathcal{H}^s(A) < \infty$, then for $\mathcal{H}^s$ almost all $x \in A$,*

$$\limsup_{\delta \downarrow 0} \left\{ \frac{\mathcal{H}^s(A \cap B)}{d(B)^s} : B \text{ is a closed ball with } \right. \\ \left. x \in B \text{ and } d(B) < \delta \right\} \leq 1.$$

Densities of Radon measures

The densities can of course be defined for general measures.

6.8. Definition. Let $0 \leq s < \infty$ and let μ be a measure on $\mathbf{R}^n$. The upper and lower s -densities of μ at $a \in \mathbf{R}^n$ are defined by

$$\Theta^{*s}(\mu, a) = \limsup_{r \downarrow 0} (2r)^{-s} \mu(B(a, r)),$$

$$\Theta_*^s(\mu, a) = \liminf_{r \downarrow 0} (2r)^{-s} \mu(B(a, r)).$$

If they agree, their common value

$$\Theta^s(\mu, a) = \Theta^{*s}(\mu, a) = \Theta_*^s(\mu, a)$$

is called the s -density of μ at a .

If μ is a Borel measure, the above densities are Borel functions; see Exercise 3. Information on upper s -densities can be used to compare μ with $\mathcal{H}^s$.

6.9. Theorem. Let μ be a Radon measure on $\mathbf{R}^n$, $A \subset \mathbf{R}^n$, and $0 < \lambda < \infty$.

- (1) If $\Theta^{*s}(\mu, x) \leq \lambda$ for $x \in A$, then $\mu(A) \leq 2^s \lambda \mathcal{H}^s(A)$.
- (2) If $\Theta^{*s}(\mu, x) \geq \lambda$ for $x \in A$, then $\mu(A) \geq \lambda \mathcal{H}^s(A)$.

This can be proven with similar arguments to Theorem 6.2. To prove (2), consider open sets V with $A \subset V$ and the Borel sets $\{x \in V : \Theta^{*s}(\mu, x) \geq \lambda\}$ which contain A . On the other hand, Theorem 6.2 (1) follows from Theorem 6.9 when applied to $\mu = \mathcal{H}^s \llcorner A$.

Density theorems for packing measures

For packing measures the lower densities are more useful than the upper densities.

6.10. Theorem. Suppose $A \subset \mathbf{R}^n$ with $\mathcal{P}^s(A) < \infty$. Then

$$\Theta_*^s(\mathcal{P}^s \llcorner A, x) = 1 \quad \text{for } \mathcal{P}^s \text{ almost all } x \in A.$$

Proof. Recalling Exercise 1.2 and using the Borel regularity of $\mathcal{P}^s$ we may assume that A is a Borel set. Then $\mathcal{P}^s \llcorner A$ is a Radon measure by Theorem 1.9 (2) and Corollary 1.11. To prove that the lower density is

at least one almost everywhere in A , it suffices to show that for any t , $0 < t < 1$, the set

$$B = \{x \in A : \Theta_*^s(\mathcal{P}^s \llcorner A, x) < t\}$$

has zero $\mathcal{P}^s$ measure. Let $E \subset B$ and $\varepsilon > 0$. There is $\delta > 0$ such that $P_\delta^s(E) \leq \mathcal{P}^s(E) + \varepsilon$. Applying Theorem 2.8 we find disjoint balls $B_i = B(x_i, r_i)$, $i = 1, 2, \dots$, such that $d(B_i) < \delta$, $x_i \in E$,

$$\mathcal{P}^s(A \cap B_i) < t d(B_i)^s$$

and

$$\mathcal{P}^s\left(E \setminus \bigcup_i B_i\right) = 0.$$

Then

$$\begin{aligned} \mathcal{P}^s(E) &\leq \sum_i \mathcal{P}^s(E \cap B_i) \leq \sum_i \mathcal{P}^s(A \cap B_i) \\ &< t \sum_i d(B_i)^s \leq t P_\delta^s(E) \leq t(\mathcal{P}^s(E) + \varepsilon). \end{aligned}$$

Letting $\varepsilon \downarrow 0$, we have $\mathcal{P}^s(E) \leq t \mathcal{P}^s(E)$ for $E \subset B$. Therefore, whenever $B = \bigcup_i E_i$,

$$\mathcal{P}^s(B) \leq \sum_i \mathcal{P}^s(E_i) \leq t \sum_i \mathcal{P}^s(E_i),$$

which implies $\mathcal{P}^s(B) \leq t \mathcal{P}^s(B)$ and so $\mathcal{P}^s(B) = 0$.

To prove the opposite inequality, let $t > 1$, $r_0 > 0$ and

$$B = \{x \in A : \mathcal{P}^s(A \cap B(x, r)) \geq t(2r)^s \text{ for } 0 < r < r_0\}.$$

It is enough to show $\mathcal{P}^s(B) = 0$. One easily sees that B is a Borel set, recall 2.10.

Let $\varepsilon > 0$. Due to Theorem 1.10 applied to $\mathcal{P}^s \llcorner A$ there are a compact set F and an open set U such that $F \subset B \subset U$ and $\mathcal{P}^s(A \cap U) < \mathcal{P}^s(B) + \varepsilon < \mathcal{P}^s(F) + 2\varepsilon$. Let $0 < \delta < \min\{r_0, d(F, \mathbf{R}^n \setminus U)\}$. If $B_1, B_2, \dots$ are disjoint closed balls with centres in F and with $d(B_i) \leq \delta$, then $B_i \subset U$ and

$$t \sum_i d(B_i)^s \leq \sum_i \mathcal{P}^s(A \cap B_i) \leq \mathcal{P}^s(A \cap U) < \mathcal{P}^s(B) + \varepsilon.$$

Thus $t P_\delta^s(F) \leq \mathcal{P}^s(B) + \varepsilon$ and, letting δ tend to zero, $t \mathcal{P}^s(F) \leq \mathcal{P}^s(B) + \varepsilon$. Consequently,

$$t \mathcal{P}^s(B) \leq t(\mathcal{P}^s(F) + \varepsilon) \leq t(\mathcal{P}^s(B) + \varepsilon) \leq \mathcal{P}^s(B) + (1+t)\varepsilon.$$

Letting $\varepsilon \downarrow 0$, we have $t \mathcal{P}^s(B) \leq \mathcal{P}^s(B) < \infty$, which gives $\mathcal{P}^s(B) = 0$ as $t > 1$. $\square$

As in 6.9 similar proofs give

6.11. Theorem. Let μ be a Radon measure on $\mathbf{R}^n$, $A \subset \mathbf{R}^n$ and $0 < \lambda < \infty$.

- (1) If $\Theta_*^s(\mu, x) \leq \lambda$ for $x \in A$, then $\mu(A) \leq \lambda \mathcal{P}^s(A)$.
- (2) If $\Theta_*^s(\mu, x) \geq \lambda$ for $x \in A$, then $\mu(A) \geq \lambda \mathcal{P}^s(A)$.

Combining the information we have about the Hausdorff and packing measures, we have

6.12. Theorem. Let $A \subset \mathbf{R}^n$ with $\mathcal{P}^s(A) < \infty$. Then $\mathcal{P}^s(A) = \mathcal{H}^s(A)$ if and only if the density $\Theta^s(A, x)$ exists and equals 1 for $\mathcal{P}^s$ almost all $x \in A$.

Proof. Assume $\mathcal{P}^s(A) = \mathcal{H}^s(A)$. Using the Borel regularity and Exercise 1.2, we may assume that A is a Borel set. Then, as $\mathcal{H}^s \leq \mathcal{P}^s$ by Theorem 5.12, we have $\mathcal{P}^s(B) = \mathcal{H}^s(B)$ for all Borel sets $B \subset A$. Hence by Theorems 6.10 and 6.2 (1) for $\mathcal{H}^s$ almost all $x \in A$,

$$1 = \Theta_*^s(\mathcal{P}^s \llcorner A, x) = \Theta_*^s(A, x) \leq \Theta^{*s}(A, x) \leq 1.$$

Thus $\Theta^s(A, x) = 1$ for $\mathcal{H}^s$ almost all $x \in A$. Since $\mathcal{H}^s$ and $\mathcal{P}^s$ agree for Borel subsets of A , they agree for all subsets of A by the Borel regularity. Hence $\Theta^s(A, x) = 1$ for $\mathcal{P}^s$ almost all $x \in A$.

Suppose $\Theta^s(A, x) = 1$ for $\mathcal{P}^s$ almost all $x \in A$. Let B be a Borel set with $A \subset B$ and $\mathcal{H}^s(A) = \mathcal{H}^s(B)$. We have $\Theta_*^s(B, x) = \Theta_*^s(A, x)$ for $x \in \mathbf{R}^n$ by Exercise 1.2 and Theorem 5.12 (which gives $\mathcal{H}^s(A) < \infty$). Then by Theorem 6.11 (2) applied to the Radon measure $\mu = \mathcal{H}^s \llcorner B$,

$$\begin{aligned} \mathcal{H}^s(A) &\geq \mathcal{H}^s(\{x \in A : \Theta_*^s(B, x) \geq 1\}) \\ &\geq \mathcal{P}^s(\{x \in A : \Theta_*^s(A, x) \geq 1\}) = \mathcal{P}^s(A). \end{aligned}$$

Thus $\mathcal{H}^s(A) = \mathcal{P}^s(A)$ by Theorem 5.12. □

In Chapters 14 and 17 we shall see that the condition $\Theta^s(A, x) = 1$ for $\mathcal{H}^s$ almost all $x \in A$ (which holds by the above theorem if $\mathcal{P}^s(A) = \mathcal{H}^s(A) < \infty$) brings strong restrictions. It implies that s must be an integer, if $\mathcal{H}^s(A) > 0$, and A rather regular; $\mathcal{H}^s$ almost all of it can be covered with countably many s -dimensional C^1 submanifolds.

In terms of the lower densities we get the following sufficient condition for the Hausdorff and upper packing dimensions to agree.

6.13. Theorem. Let $A \subset \mathbf{R}^n$ with $0 < \mathcal{H}^s(A) < \infty$. If $\Theta_*^s(A, x) > 0$ for $\mathcal{P}^s$ almost all $x \in A$, then $\overline{\dim}_p A = \dim A$.

Proof. We assume that A is a Borel set and leave the modification of the usual type to the reader. As noted in 5.9, $s = \dim A \leq \overline{\dim}_p A$. Let $B = \{x \in A : \Theta_*^s(A, x) = 0\}$ and $C = \{x \in A : \Theta^{*s}(A, x) > 1\}$. Then $\mathcal{P}^s(B) = 0$. By Theorem 6.2 (1), $\mathcal{H}^s(C) = 0$, which by Theorem 6.11 (2) implies $\mathcal{P}^s(C \setminus B) = 0$, whence $\mathcal{P}^s(B \cup C) = 0$. We can write $A \setminus (B \cup C) = \bigcup_{i=1}^{\infty} A_i$ where the sets A_i are bounded, $\mathcal{H}^s(A_i) > 0$, and with some $r_i > 0$, $0 < a_i < b_i < \infty$,

$$a_i r^s \leq \mathcal{H}^s(A \cap B(x, r)) \leq b_i r^s \quad \text{for } x \in A_i, \quad 0 < r < r_i.$$

Then $\overline{\dim}_p(B \cup C) \leq s$ by Theorem 5.11, and for $i \geq 1$, $\overline{\dim}_M A_i = \dim A_i = s$ by Theorem 5.7, and so by 5.9 $\overline{\dim}_p A_i = s$. Consequently, $\overline{\dim}_p A = s$. $\square$

Remarks related to densities

6.14. (1) Recently there has been a great interest in the multifractal structure of Borel measures μ on $\mathbf{R}^n$. For $0 < \alpha < \infty$ let A_α be the set of those x for which $\mu(B(x, r))$ behaves like r^α for small r . Usually this means more precisely that

$$A_\alpha = \left\{ x : \lim_{r \downarrow 0} \frac{\log \mu(B(x, r))}{\log r} = \alpha \right\}.$$

The problem is what one can say about $\dim A_\alpha$. For general Borel measures there is very little to say but it has turned out that if μ is for example in some sense self-similar, then $\dim A_\alpha$ can be computed from a simple formula. For such results see e.g. Brown, Michon and Peyrière [1], Cawley and Mauldin [1], Deliu, Geronimo, Shonkwiler and Hardin [1], Edgar and Mauldin [1], Falconer [16], [23], Fan [2], Geronimo and Hardin [1], Jaffard [1], King and Geronimo [1], Lopes [1]–[2], Olsen [1]–[2], Pesin [1]–[2], Peyrière [3] and Rand [1].

(2) Somewhat related to the above are the works of Cutler [1] and Kahane and Katznelson [1] who disintegrate a general measure μ into its α -dimensional components, see also Cutler [2]–[3] and Cutler and Dawson [1]. Cutler works directly with the Hausdorff dimension whereas Kahane and Katznelson use potential-theoretic concepts (cf. Chapter 8).

(3) For the Cantor sets $C(\lambda)$ of 4.10,

$$\Theta_*^s(C(\lambda), x) < \Theta^{*s}(C(\lambda), x) \quad \text{for } x \in C(\lambda)$$

with $s = \dim C(\lambda)$ so that the density does not exist. However, Bedford and Fisher [1] found that a different average density does exist. That is, they show that for $E = C(1/3)$, and also for many other sets, with $s = \dim E$, the limit

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T e^{ts} \mathcal{H}^s(E \cap B(x, e^{-t})) dt$$

exists and equals a constant for $\mathcal{H}^s$ almost all $x \in E$. In fact, by the results of Falconer [20] and Patzschke and M. Zähle [4] this holds more generally for self-similar and even self-conformal sets in $\mathbf{R}^n$ when the open set condition holds. The proofs of these results are ergodic-theoretic in nature. For related results, see also Bedford and Kamae [1], Falconer and Springer [1], Graf [3], Patzschke and M. Zähle [1] and [3].

Exercises.

1. Show that for the Cantor sets $C(\lambda)$ of 4.10, $0 < \Theta_*^s(C(\lambda), x) < \Theta^{*s}(C(\lambda), x) < \infty$ for $x \in C(\lambda)$ with $s = \dim C(\lambda)$.
2. Use the set described at the end of 4.12 to give an example of a compact set $A \subset \mathbf{R}^n$ such that $0 < \mathcal{H}^s(A) < \infty$ and $\Theta_*^s(A, x) = 0$ for $x \in A$.
3. Prove that for a Borel measure μ the densities $\Theta^{*s}(\mu, \cdot)$ and $\Theta_*^s(\mu, \cdot)$ defined in 6.8 are Borel functions. Hint: Recall 2.10.
4. Let μ be a Radon measure on $\mathbf{R}^n$. Prove that $\mu \ll \mathcal{H}^s$ if and only if $\Theta^{*s}(\mu, x) < \infty$ for μ almost all $x \in \mathbf{R}^n$.
5. Let μ be a Radon measure on $\mathbf{R}^n$. Prove that $\mu \ll \mathcal{P}^s$ if and only if $\Theta_*^s(\mu, x) < \infty$ for μ almost all $x \in \mathbf{R}^n$.
6. Let μ be a Radon measure on $\mathbf{R}^n$ and $t \geq 0$. Show that

$$\dim \left\{ x : \liminf_{r \downarrow 0} \frac{\log \mu(B(x, r))}{\log r} \leq t \right\} \leq t$$

and

$$\overline{\dim}_p \left\{ x : \limsup_{r \downarrow 0} \frac{\log \mu(B(x, r))}{\log r} \leq t \right\} \leq t.$$

7. Let μ be a Radon measure on $\mathbf{R}^n$ and $t \geq 0$. Show that if A and B are subsets of $\mathbf{R}^n$ with $\mu(A) > 0$, $\mu(B) > 0$,

$$\liminf_{r \downarrow 0} \frac{\log \mu(B(x, r))}{\log r} \geq t \quad \text{for } x \in A$$

and

$$\limsup_{r \downarrow 0} \frac{\log \mu(B(x, r))}{\log r} \geq t \quad \text{for } x \in B,$$

then $\dim A \geq t$ and $\overline{\dim}_p B \geq t$.